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Cover Letter

Dear Members of the Faculty Search Committee,

I am writing to express my interest in tenure-track faculty opportunities in Industrial Engineering, Systems Engineering, Manufacturing Engineering, or closely related areas where data science, artificial intelligence, and rigorous statistical modeling are used to address complex engineered systems. I am a Ph.D. Candidate in Industrial Engineering at Arizona State University, advised by Dr. Ashif Iquebal, and I expect to complete my Ph.D. in Spring 2027. My research develops physics-aware and uncertainty-aware computational methods for inverse problems, governing-physics discovery, and rapid engineering prediction. Across this work, I combine Bayesian inference, stochastic modeling, machine learning, operator learning, and first-principles simulation to build models that remain useful when measurements are sparse, physical conditions vary, and high-fidelity computation is expensive.

My doctoral research has produced three complementary methodological foundations for an independent faculty program. First, my work on Bayesian spatiotemporal inverse heat conduction, published in the ASME Journal of Heat and Mass Transfer, develops a statistically principled approach for recovering hidden physical quantities from sparse and uncertain measurements. I have extended stochastic inverse modeling to non-destructive characterization of elastoplastic material properties from indentation data, with an ongoing journal manuscript involving collaborators at the National Institute of Standards and Technology and Oak Ridge National Laboratory. Second, my 2026 Communications Physics article, “Discovering Governing Physics in the Presence of Uncertainty,” introduces a stochastic inverse physics-discovery framework that treats unknown governing-equation coefficients as random variables. Across ten benchmark and real-data scenarios, the method reduced coefficient error by about 80% relative to seven comparison approaches and achieved exact structural identification in seven cases. Third, CRONet, published in Manufacturing Letters, uses operator learning to accelerate topology optimization by learning a surrogate for repeated finite-element calculations. Together, these projects define a coherent trajectory: infer what cannot be observed directly, identify what physics is missing, and accelerate predictive models so that they can support repeated design and decision making.

As a faculty member, I plan to develop this trajectory into a broader research program in uncertainty-aware scientific machine learning for engineering systems. One thrust will create Bayesian and data-consistent models that integrate heterogeneous sensor streams while distinguishing process variability, measurement error, model discrepancy, and surrogate uncertainty. A second will develop self-correcting models that use stochastic physics discovery and active experimentation when observations are no longer explained by the assumed governing model. A third will extend neural-operator surrogates to coupled, higher-dimensional engineering problems so that inference, optimization, and control can be performed at decision-relevant time scales. Advanced manufacturing and materials characterization will remain important proving grounds, while the methodological contributions are intended to transfer to semiconductor, aerospace, thermal, energy, and other complex engineered systems. This agenda is designed to support competitive early-career funding and collaborative research with national laboratories, industry, and researchers across engineering and data science.

Teaching and mentoring are central to how I envision this program. My teaching experience spans mathematics and physics instruction in Nigeria and Industrial Engineering courses at Arizona State University, including Probability and Statistics for Engineering Problem Solving and Advanced Quality Control. I organize instruction around a simple goal: students should be able to choose an appropriate model, explain its assumptions, test whether those assumptions are reasonable, and translate the output into a defensible engineering decision. I use worked examples, computational notebooks, staged projects, and authentic engineering data to make abstract statistical and machine-learning concepts observable. I am prepared to contribute to courses in probability and statistics, quality and reliability, data analytics, simulation, machine learning, manufacturing systems, and systems modeling. I would also develop graduate instruction in physics-informed machine learning, uncertainty quantification, and digital twins, and an undergraduate course in manufacturing data analytics built around sensing, Python-based analysis, and project-based learning.

My mentoring approach emphasizes transparent expectations, reproducible computational practice, and increasing intellectual ownership. I have mentored undergraduate researchers through ASU summer programs on machine learning for CAD model generation and active learning, and I plan to build tiered research teams in which experienced graduate students mentor junior researchers while receiving explicit training in supervision and research communication. I also intend to create remote and hybrid research opportunities that make meaningful research participation possible for students who may have limited access to research-intensive environments. In the classroom, transparent rubrics, iterative feedback, multiple modes of technical communication, and deliberately structured team roles help students with different preparation levels participate while preserving common technical standards.

I also value professional service and collaborative problem solving. My service has included leadership in the ASU INFORMS Student Chapter, conference session organization and chairing, peer review, and STEM outreach. In 2026, I contributed to interdisciplinary engineering challenges spanning manufacturing data and semiconductor systems. Team RAV received First Place and the Best Presentation Design Award in the NSF Future Manufacturing Data Challenge and was also recognized as a winning team in the Semiconductor Solutions Challenge for a machine-learning approach to static voltage-drop prediction. These experiences have reinforced my interest in team-based, sponsor-relevant research and in creating opportunities that connect students with industrial and research partners.

I am seeking a faculty environment that values rigorous methodological research, interdisciplinary collaboration, engaged teaching, and partnerships that move research from theory to consequential engineering applications. I would bring a research agenda at the intersection of Bayesian inference, scientific machine learning, uncertainty quantification, and engineering systems, together with a commitment to mentoring students as independent problem solvers. My curriculum vitae, research statement, and teaching statement provide additional detail on these activities and future plans. Thank you for your consideration. I would welcome the opportunity to discuss how my research and teaching can contribute to the goals of a forward-looking engineering department.

Sincerely,
Ridwan O. Olabiyi